

John Misiko Kakai
Data Scientist and AI Engineer

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PROFESSIONAL SUMMARY.

- Data Scientist, AI Engineer, and Data Engineer with 5+ years of experience, including work with UN OCHA and UNFPA, building data-driven solutions across humanitarian and development contexts.
- Skilled in Python, R, SQL, data engineering, machine learning, AI, LLMs, RAG, geospatial analytics, and cloud platforms, with experience developing end-to-end data and AI systems across AWS, Azure, and GCP.
- Impact-driven and analytical professional who combines technical expertise, problem-solving, and clear communication to transform complex data into actionable insights that support better decisions and real-world outcomes.

EDUCATION

Carnegie Mellon University, Pittsburgh, USA
MSc, Information Technology (Data Science and Machine Learning) May 2026
Mastercard Foundation Scholar GPA: 3.71/4.00

Relevant Coursework: Machine Learning in Production, Deep Learning, Data Engineering, Applied Machine Learning, Software Architecture, Machine Learning for Earth Observation, Cloud Computing, Computational DS.

University of Nairobi Nairobi, Kenya
BSc, Geospatial Engineering, Second Class Honours, Upper Division 2021

Relevant Coursework: Statistics and Probability, Programming and Numerical Methods, Database Systems, Remote Sensing and Image Analysis, Geographic Information Systems, Spatial Data Analysis

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL, R (basic), Bash

Data Engineering and Management: Apache Spark, AWS Glue, Airflow concepts, ETL and ELT pipelines, REST APIs, PostgreSQL, OpenSearch, data modelling, data integration, data quality

Data Science and Statistics: Statistical analysis, predictive modelling, time series forecasting, anomaly detection, clustering, hypothesis testing, survey data analysis, prevalence estimation, indicator development, feature engineering, EDA

Machine Learning and Deep Learning: scikit-learn, XGBoost, Random Forest, PyTorch, TensorFlow/Keras, CNN, LSTM, ResNet, BERT, HuggingFace Transformers, hyperparameter tuning

LLMs and Generative AI: LangChain, OpenAI API, Anthropic API, RAG architectures, ChromaDB, FAISS, embeddings, prompt engineering, fine-tuning, evaluation

Health, Survey and Monitoring Systems: DHIS2, MICS, health information systems, monitoring and evaluation, results-based management

Digital Data Collection: ODK, KoboToolbox, survey methodology, field data collection, surveillance systems

MLOps and Full Lifecycle: MLflow, DVC, Docker, GitHub Actions, ONNX, model deployment, observability, drift detection, automated retraining, A/B rollout, model maintenance

Cloud Platforms: AWS (Glue, S3, RDS, OpenSearch, Lambda), Azure (ML, AI Foundry), GCP (Vertex AI, BigQuery, Cloud Storage)

Geospatial AI: Google Earth Engine (GEE), ArcGIS, QGIS, geospatial data integration, spatial analysis

Visualisation and BI: Plotly, Matplotlib, Seaborn, Streamlit, interactive dashboards, executive-ready reporting

Collaboration: Git, GitHub, Agile workflows, cross-functional teams, technical writing

RELEVANT EXPERIENCE

United Nations Population Fund (UNFPA) Nairobi, Kenya
Data Analyst Specialist Sep 2026 to Present

- Conduct large-scale statistical and data analysis across multi-country datasets to generate FGM risk, attitudinal incidence, and subnational prevalence estimates using DHIS, MICS, and other survey data..

- Integrate programmatic, geospatial, social media, and health information data to strengthen evidence-based monitoring and results-based management across the Joint Programme on the Elimination of Female Genital Mutilation.
- Develop and manage data pipelines and analytical workflows supporting the integration of historical FGM social media data into UNFPA's Demographic Intelligence from Open Sources (DIOS) platform.
- Support pilot testing of social norm indicators within health surveillance systems using digital data-collection platforms including ODK and KoboToolbox.
- Apply Python, SQL, geospatial analytics, and AI-driven methods to link programmatic and epidemiological data and generate actionable insights across multiple countries

United Nations Office for the Coordination of Humanitarian Affairs
Humanitarian Data Science Intern

Geneva, Switzerland
 June 2026 to Aug 2026

- Supported the implementation and enhancement of the HPC Country Module, facilitating partner reporting (5W), data collection, and humanitarian planning processes; led the preparation and dissemination of ActivityInfo training videos to support partner onboarding and effective use of the platform.
- Applied data analytics and visualization techniques to support humanitarian needs assessments, response planning, monitoring, and reporting.
- Utilized Python, NumPy, and related analytical tools to process, analyze, and communicate humanitarian data for evidence-based decision-making.
- Developed dashboards, infographics, and data-driven insights to strengthen humanitarian coordination, strategic prioritization, and emergency response operations.
- Executed machine learning, data science, and geospatial analytics projects and technical tasks assigned by the direct supervisor [Mr Bassil Aleter](#), delivering data-driven solutions to support humanitarian programs and operational objectives.

Bite Data

Data Scientist

USA, Remote
 Jan 2026 to June 2026

- Designed and optimised AWS data pipelines (Glue, S3, RDS, OpenSearch) and built scalable Python, SQL, and Apache Spark ETL workflows to transform raw trade, customs, and compliance datasets into analytics ready data products
- Developed predictive models for anomaly detection, sanctions risk scoring, trade forecasting, and supply chain intelligence; tuned hyperparameters and evaluated against business metrics.
- Designed and developed LLM-powered AI agents for customs and trade intelligence, combining RAG, structured data tools and prompt engineering to automate data exploration, compliance analysis, insight generation, and decision-support workflows.

Carnegie Mellon University

Graduate Research Assistant, AI and ML Systems (Mobility Laboratory)

Pittsburgh
 October 2024 to February 2025

- Owned end to end ML lifecycle on Azure ML: data preparation, training, hyperparameter tuning, containerised (Docker) deployment, CI/CD with GitHub Actions, observability dashboards, drift detection, automated retraining, and model maintenance
- Built a GPT 4o powered RAG system using LangChain and ChromaDB with hybrid dense and sparse retrieval, chunking strategies, and prompt engineering for institutional knowledge search
- Fine-tuned BERT models for question answering (SQuAD) and sentiment analysis on Azure AI Foundry; applied SHAP based explainability and responsible AI documentation
- Designed and trained a hybrid CNN LSTM deep learning model for spatio temporal prediction; first author paper accepted at IEEE MIGARS 2025.

Spatialise

ML Engineer, Data Pipelines

Netherlands (Remote)
 June 2024 to September 2024

- Designed and maintained production grade ML pipelines using Python and Git based collaborative workflows; implemented automated preprocessing, feature extraction, and anomaly detection
- Built and benchmarked ML models (Random Forest, CNN, XGBoost) for classification and change detection; documented training, evaluation, and tuning in Jupyter notebooks for reproducibility

- Built data analytics workflows that combined field surveys, GIS layers, and operational datasets into clean, analysis ready inputs for telecommunications infrastructure planning
- Produced data driven technical reports and visualisations that informed Safaricom FTTX and FTTH network design decisions for senior planning and engineering leadership
- Led client meetings and cross functional coordination to secure timely approvals for designs informed by coverage and demand analysis

PROJECT EXPERIENCE

Azure ML Production Deployment and Monitoring System, Full ML Lifecycle 2025

- Implemented end to end ML deployment on Azure ML covering MLflow model registry, Docker containerisation, automated data drift detection, performance dashboards, prediction logging, alerting, and scheduled retraining
- Designed observability and maintenance workflows to keep models reliable in production, with rollback paths and clear evaluation criteria for new model versions

Alzheimer's Disease Classification Using Deep Learning | 2025 2025

- Developed a ResNet-50 model to classify Alzheimer's disease severity from MRI scans across four disease categories, achieving 98.13% accuracy.
- Applied data augmentation, hyperparameter tuning, and Integrated Gradients for model optimisation and interpretability.

LLM Powered RAG Document Retrieval Agent, Applied AI 2026

- Built a production RAG system using GPT 4o, LangChain, ChromaDB, and HuggingFace embeddings with chunking strategies, hybrid dense and sparse retrieval, and prompt engineering for technical Q and A
- Engineered evaluation harnesses and feedback loops to measure retrieval quality and response faithfulness, informing iterative tuning

Spatio Temporal Deep Learning System for Soil Moisture Mapping IEEE MIGARS 2025 (First Author, Accepted) 2024 to 2025

- Designed and trained a hybrid CNN LSTM model integrating multi sensor data for time series prediction; built the full Python pipeline from preprocessing and feature engineering to training, hyperparameter tuning, and inference

Additional projects: [GitHub](#)

ADDITIONAL EXPERIENCE

GeoSight AI, Founder Nairobi, Kenya *AI Powered Decision Support Platform* 2025 to Present

- Building an AI driven decision support platform for smallholder cooperatives, combining data pipelines, ML models, and analytics; selected for FAST Program Cohort 3 (Inkomoko) Build Phase grant interview

Certifications and Recognition

- Mastercard Foundation Scholar, Carnegie Mellon University;
- Microsoft Azure AI and ML platform experience (Azure ML Studio, Azure AI Foundry compatible workflows);
- Graduate Member, Institution of Surveyors of Kenya